From Quantitative Indexes' Stability to that of Stylometry: A Homogeneous Texts Approach to EFL Academic Writing

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ABSTRACT. We begin to verify the hypothesis that EFL teachers who have published academic writing have a stable writing style for homogeneous written texts, as revealed by the stability of quantitative indexes. To achieve this, we analyzed ten homogeneous academic texts from an EFL academic monograph and 26 homogeneous academic texts from the academic papers of three EFL teachers, calculating quantitative indexes using the Quantitative Index Text Analyser (QUITA) software. Pearson correlation and principal component analysis were used to analyze the major quantitative indices of writing styles. The results show that all the principal quantitative indices (excepting verb distances) maintain a high degree of stability, reflecting the stable writing style of the author; (1) Writer's view, activity, and thematic concentration; and (2) entropy, descriptivity, and thematic concentration can represent the individual's writing style. The verified hypothesis can be further applied to help identify the EFL written text authorship and contribute to academic writing ethics education.In this way, we suggest that self-checking policy in higher education be adopted to promote education development before publishing academic writing.

Keywords: QUITA; Quantitative indexes' stability; Homogeneous academic written texts; Authorship identification; Ethics education

1. Introduction

Universities or colleges, traditionally places for higher education and basic research, are also ranked for academic and research performance [1]. English as a foreign language (EFL) or second language (ESL) academic writing is an expected behavior in higher education. Different EFL academic writers are supposed to show varied writing styles with individual differences. In the internet age, with a few clicks of the mouse, students can 'cut and paste' the information they need [2]. Most students believe that it is acceptable to use some internet materials without proper citations and referencing, which also contributes to the dilution of individual differences among writers. Ethics in education can play a moral and influential role [3]. Scientific and academic misconduct is a severe problem in higher education, and we should strengthen academic ethics education. Ethics should be defined as the principles by which people act in a knowledge context, based on what they know, ought to know, and should know and how to do [4]. The question is where and how to engage stakeholders in ethics education to reduce academic written misconduct. It should be a concern for all countries and regions.

China has thousands of English as a foreign language (EFL) learners. Most individuals are required to publish academic papers and theses to graduate or advance in their careers. Correspondingly, academic misconduct inevitably arises. A disturbingly large number of plagiarism cases have been detected in recent years [5]. To identify misconduct in an article or book, in essence, is to determine whether the author involved maintains a consistent, stable writing style despite individual differences. To be more exact, it concerns the author's attribution. The issue of authorship attribution warrants greater research attention, particularly in light of the increasing number of unknown English academic written texts. It should be the focus of academic ethics education. In short, EFL writers should increase their consciousness of author identification when writing academic papers or theses through ethics education.

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Authorship attribution, also called authorship identification, refers to recognizing the author of the anonymous or unknown text by analyzing the target text's stylometry or writing style. The main idea is to show the author's unconscious writing habits in the text through some quantifiable features and then highlight the literary features or writing style of the work to determine the author of the anonymous text [6].

The traditional research on authorship attribution is predominantly based on empirical knowledge from literature or linguistics, partly because the degree of socialization in scientific research is relatively low, and the application of mathematics and other related tools was not widespread before the 19th century [7]. During this period, Western researchers inferred the authorship of unknown sonnets based on changes in rhythm and rhyme. Since the 19^{th} century, research has gradually evolved into a new landscape by incorporating more mathematical tools. Mendenhall (1887) was the first researcher to apply mathematical tools to quantify a writer's writing style by using a unique curve to attribute authorship to Shakespeare's plays and other works [8]. It marks the beginning of modern research on authorship attribution. Computer technology and the internet, developed in the 1990s, significantly contributed to the advancement of computational stylometry. Nowadays, the target of unknown or anonymous texts focuses on literature (e.g., Digamberrao & Prasad, 2018 [9]; Hoover, 2004 [10]; Zhao & Zobel, 2007 [11]) and messages (e.g., Boutwell, 2011 [12]; Li et al., 2014 [13]; Schwartz et al., 2013 [14]; Srinivasan & Nalini, 2019 [15]).

Authorship attribution is a field related to multiple disciplines. To simplify the problem and facilitate research, researchers typically propose hypotheses. The first hypothesis is that the author's writing style is influenced by their characteristics, such as status, gender, personality, age, and educational level [16]. It is vital to research authorship attribution. The second hypothesis is that these characteristics of the author can be observed in their writing style [17]. This hypothesis is the most important one in the study of author identification. This hypothesis provides a theoretical basis for quantifying the author's writing style. However, the author's writing style could be more consistent. It will be influenced by various external factors, including social background, time, theme of the text, media, and audience.

Meanwhile, researchers also agree that invariant properties of textual statistics exist [18]. The existence of such invariants implied the likelihood that some related feature might be found that was at least invariant for any given author, though possibly varying among different authors [19]. Besides, regardless of what these invariants are, they can be quantified by researchers through various means. The researchers must maintain the textual properties as invariant as possible, rather than separating them, because this will minimize certain classification loss functions [19].

Stylometry refers to the analytic study of literary styles, mainly applied to questions of authorship. The author's writing style stems from their thinking and expression. In the process of expression, the author unconsciously integrates their personality and social background. Although the author's writing style will gradually change, researchers typically assume that invariant properties of writing style always exist and can be quantified through technical means. The literary style's stylistic features are divided into unitary invariant and multivariate. The features of unitary invariants mainly include word length, sentence length, paragraph number, and total vocabulary. Multivariate invariants are often further combinations of some unitary invariants. Research indicates that multivariate features can significantly enhance the accuracy of author recognition [20]. According to the demand and complexity of stylistic features for linguistic calculation, the multivariate features can be divided into character, lexical, syntactic, and semantic features [6]. All the features can be calculated through quantitative text indexes.

Stylometry can be assessed through text analysis, where text typically refers to a passage, extract, or complete piece of writing or speech [21]. Text is part of language, and language is the carrier of culture and thinking. Native writers or second/foreign language writers can write target texts related to writing style analysis. English, a world language, is preferred by most researchers when choosing a text. While few researchers have addressed the question of what the relationship is between a native writer and a second/foreign language writer in terms of text stylometry?

Nevertheless, English as a second/foreign language, ESL/EFL writing, is a cross-cultural communicative behavior that transforms thinking and cultural models [22]. Most authors of current target texts who are concerned with writing style analysis are English natives, and the text types are primarily literary and informational. So, what is the landscape of the EFL academic texts in terms of stylometry?

2. Research Hypothesis and Questions

Writing an academic paper or thesis in English can be a challenging task, even for native speakers. It is significantly more challenging for EFL learners to accomplish the same task. In most cases, EFL teachers who can write a qualified English academic paper or thesis have acquired their English writing skills over many years. Based on this, the authors put forward the research hypothesis: The writing style of EFL homogeneous academic texts remains relatively stable in terms of the stability of the quantitative indices. In other words, EFL/ESL academic writers should embody individual differences with different personalities.

We defined *homogeneous texts* as texts with the following attributes: (1) the same genre of texts, (2) from the same author, (3) of similar length, (4) written within a specific period, and (5) with similar topics. Likewise, EFL homogeneous academic texts are similar in topic to academic writings of the same length from the same EFL author within a specific period.

The research hypothesis considered, this study focuses on the following three research questions:

- Can the quantitative indices of the EFL homogeneous academic written texts remain stable? Why or why not?
- Which quantitative indices can represent the writing style of the same author's EFL homogeneous academic written texts?
- How can the research results be applied to identify authorship and contribute to academic writing ethics education in a university?

3. Methods

3.1. Instrument

We measured all the texts in the research using *the Quantitative Index Text Analyzer* (QUITA) software, a free software developed by linguistics faculty and students at Palacky University, the Czech Republic. QUITA is available for download at http://oltk.upol.cz/software or https://kcj.osu.cz/wp-content/uploads/2018/06/QUITA_Setup_1190.zip.

3.2. Quantitative Text Indexes

Some quantitative text indexes are necessary to assess the stability of the writing style in the target texts. All indexes used in this research come from Haitao Liu's *An Introduction to Quantitative Linguistics* (2017)[23]. Liu briefly introduces twenty-two quantitative text indexes in his book. In this research, however, we only applied 12 of the most frequently used indexes, including TTR, h-point, entropy, R_1, RR, RR_{mc}, TC, secondary TC, activity, descriptivity, writer's view, and verb distances. Table 1 below shows the relevant details for the twelve indices.

Table 1. Brief information on the twelve quantitative indexes of text analysis

Quantitative indexes	Brief introduction of the principal quantitative indices
TTR	Token/type ratio: The ratio between the total number of word types （V）and that of word tokens （N） [24][25]and an index on lexical diversity [26].
h-point	Refers to one of the thresholds of word rank-frequency distribution and reflects lexical richness [27].
entropy	Refers to the probability of the word emerging in the text [23][28].
R_1	Relates to the richness of vocabulary, especially the richness of made-up words [29].
RR	Refers to the repeat rate based on the probability of a word emerging [23][28].
RRmc	The relative repeat rate, which can be a standardized process of RR [23][28].
TC	Describes the thematic concentration of the target text [23][28].
STC	Describes the secondary thematic concentration in case no TC is found in the text [23][28].
activity	Refers to the degree of activity in the form of verbs/verbs + adjectives [23][28].
descriptivity	Refers to the degree of descriptivity in the form of adjectives/verbs + adjectives [23][28].
writer’s view	Refers to the degree of the triangle consisting of the h-point, the starting point of word rank-frequency distribution, and that of the ending point, which reflects the writer’s control over function words and content words in their production processes [29][30][31].
verb distances	Implies the distance between the two neighboring verbs in a sentence or neighboring sentences [23][28].

3.3. EFL Homogeneous Academic Texts

In this study, we divided EFL homogeneous academic texts into two parts. The first part originated from *Madison Pictures and American Culture*, written by author 1 (2018), a seasoned EFL teacher. His book has four parts, and the ten academic texts we used comprise the main contents of its first part (“Introduction”). Table 2 below shows additional information on the ten homogeneous academic texts.

Table 2. General information about the ten short EFL homogeneous academic texts from a book

Theme	Source	Author	Length	Genre	Token	V
Environment	Madison Pictures and American Culture	Author 1	495	Academic	489	272
Sports	Madison Pictures and American Culture	Author 1	572	Academic	571	237
Transportation	Madison Pictures and American Culture	Author 1	580	Academic	585	245
Race	Madison Pictures and American Culture	Author 1	580	Academic	595	279
Security	Madison Pictures and American Culture	Author 1	817	Academic	806	353
Education	Madison Pictures and American Culture	Author 1	490	Academic	476	258
Heroism	Madison Pictures and American Culture	Author 1	467	Academic	415	215
Religion	Madison Pictures and American Culture	Author 1	600	Academic	597	292
Pets	Madison Pictures and American Culture	Author 1	778	Academic	651	328
Welfare	Madison Pictures and American Culture	Author 1	1155	Academic	1149	442

The second portion of EFL homogeneous academic texts we used is related to academic papers by three authors: Haitao Liu, Qiufang Wen, and Yuzhi Shi, three well-known linguists in China. Table 3 below provides general information about their 13 EFL academic papers:

Table 3. General information of the 13 academic papers from three professors

Titles	Author	Sources	Years
Article 1: Quantitative Properties of English Verb Valency	Liu	Journal of Quantitative Linguistics	2011
Article 2: Dependency Direction as a Means of Word-order Typology: A Method Based on Dependency Treebanks	Liu	Lingua	(2010).
Article 3: The Complexity of Chinese Syntactic Dependency Networks	Liu	Physica A	2008
Article 4: Language Is More a Human-driven System than a Semiotic System: Comment on "Modelling Language Evolution: Examples and Predictions" by Tao Gong, Shuai Lan, Menghan Zhang	Liu	Physics of Life Reviews	2013
Article 5: Language as a Human-driven Complex Adaptive System: Comment on "Rethinking Foundations of Language from a Multidisciplinary Perspective" by T. Gong et al.	Liu	Physics of Life Reviews	2018
Article 6: Statistical Properties of Chinese Semantic Networks	Liu	Chinese Science Bulletin	2009
Article 1: On the properties of the WH-elements in Chinese	Shi	Journal of Chinese Linguistics	1997
Article 2: The establishment of the classifier system and the grammaticalization of the morphosyntactic particle de in Chinese	Shi	Language Sciences	2002
Article 1: L2 learner variables and English achievement: A study of tertiary-level English major in China	Wen	Applied Linguistics	1997
Article 2: Chinese learner corpora and second language research	Wen	International Symposium	2006
Article 3: English as a lingua franca: A pedagogical perspective	Wen	Journal of English as a Lingua Franca	2012
Article 4: The production-oriented approach to teaching university students English in China	Wen	Language Teaching	2016
Article 5: Teaching Culture (s) in English as a Lingua Franca in Asia: Dilemma and Solution	Wen	An International Conference	2015

3.4. Research Procedure

Based on the research hypothesis and questions, we conducted part-of-speech tagging using the ten English academic texts, while the twelve quantitative indices were calculated through QUITA. After tagging the ten EFL homogeneous academic texts, data for the twelve indicators were generated rapidly using QUITA.

The twelve indices from the exact text show contributions to the writing style of the same text to varying degrees, and there is the possibility of overlapping in representing the information and contribution to the text's writing style. Therefore, a Pearson correlation data analysis was performed on the calculated results of the quantitative indexes for the second step. This step was done directly through calculations in the exported Excel table and confirmed through SPSS 23. For the pair of quantitative indexes with a strong correlation or robust correlation, one of the pairs was taken as the index, forming a group for further analysis of the stability of the quantitative indexes.

Although the Pearson correlation coefficient grouping based on the calculation results of quantitative text indexes reduced the original 12 indicators to approximately 7, it still involved more quantitative indexes. Considering the unbalanced contribution of these indicators to the text's writing style and the substantial amount of work involved, a principal component analysis was performed on the quantitative indices after grouping using SPSS 23 to reduce dimensionality further.

The above steps determined the leading indicators that could reflect an author's writing style in homogeneous texts. The stability of the data was analyzed in accordance with the final selected indicators. The relevant index data were put into an Excel table to form a broken line chart. The stability of each quantitative index was obtained through qualitative data. The data analysis was used to identify the stable characteristics of the writing style by combining calculations of maximum value, minimum value, average value, variance, and standard deviation.

Given the analysis of the stability of the quantitative indices of the ten texts, the researchers performed a confirmatory analysis of the indicators using Liu's, Wen's, and Shi's 13 academic papers. Twenty-six homogeneous texts from the introductions and the conclusions of the 13 papers were calculated through QUITA without part-of-speech tagging to make the research more accessible. The stability of the text indexes for all homogeneous academic texts was intended to enhance the reliability and validity of the research.

4. Results and Discussion

4.1. Results

4.1.1. Data of the Twelve Quantitative Indexes of the Ten EFL Homogeneous Academic Texts

Table 4 below presents the data for the twelve quantitative indexes of the first author's ten EFL homogeneous academic texts, analyzed using SPSS 23.

Table 4. QUITA data of the twelve quantitative indexes of the ten homogeneous academic texts

Text	h-point	TTR	R1	entropy	RR	RRmc	writer's view	Verb distances	activity	descriptivity	TC	STC
1	7.000	0.556	0.866	7.443	0.012	0.946	1.608	0.341	0.672	0.327	0.000	0.010
2	10.000	0.415	0.802	7.036	0.015	0.937	1.604	3.048	0.774	0.226	0.028	0.017
3	9.500	0.419	0.814	7.179	0.013	0.946	1.610	23.479	0.768	0.232	0.000	0.027
4	7.500	0.469	0.801	7.292	0.015	0.932	1.599	13.235	0.643	0.356	0.000	0.019
5	11.500	0.438	0.804	7.572	0.012	0.940	1.596	12.855	0.760	0.239	0.009	0.012
6	7.500	0.542	0.858	7.391	0.011	0.953	1.621	7.333	0.545	0.454	0.025	0.047
7	8.000	0.518	0.863	7.177	0.011	0.960	1.656	18.382	0.716	0.283	0.125	0.033
8	7.000	0.489	0.828	7.487	0.012	0.948	1.611	13.284	0.529	0.470	0.000	0.006
9	9.500	0.504	0.826	7.559	0.013	0.939	1.598	4.368	0.572	0.427	0.096	0.027
10	12.000	0.385	0.737	7.728	0.012	0.933	1.588	4.105	0.655	0.344	0.071	0.016

According to Table 4, the data of the twelve quantitative indexes remain relatively stable except for the verb distances.

4.1.2. Pearson Correlation Data for the Twelve Quantitative Indexes and the Forming of the Groups

Since the twelve quantitative indexes can all contribute to the writing style of homogeneous academic texts, the Pearson correlation coefficient data needed to be calculated. For this study, data were collected using SPSS 23. Table 5 below demonstrates the results.

Table 5. Pearson correlation coefficient data matrix of the twelve quantitative indexes

	h-point	TTR	R1	entropy	RR	RRmc	writer's view	Verb distances	activity	descriptivity	TC	STC
h-point	**1**	**-0.797**	**-0.765**	0.326	0.125	-0.513	-0.504	-0.082	0.487	-0.487	0.222	-0.154
TTR	**-0.797**	**1**	**0.896**	-0.019	-0.447	**0.626**	0.515	-0.168	-0.514	0.514	0.081	0.321
R_1	**-0.765**	**0.896**	**1**	-0.355	-0.399	**0.792**	**0.697**	0.091	-0.204	0.204	0.023	0.366
entropy	0.326	-0.019	-0.355	**1**	-0.441	-0.286	-0.511	-0.357	-0.472	0.472	0.076	-0.278
RR	0.125	-0.447	-0.399	-0.441	**1**	-0.720	-0.462	-0.132	0.363	-0.363	-0.312	-0.250
RRmc	-0.513	**0.626**	**0.792**	-0.286	**-0.720**	**1**	**0.896**	0.354	-0.121	0.121	0.254	0.480
W's view	-0.504	0.515	**0.697**	-0.511	-0.462	**0.896**	**1**	0.436	0.030	-0.030	0.431	0.499
VD	-0.082	-0.168	0.091	-0.357	-0.132	0.354	0.436	**1**	0.269	-0.269	-0.056	0.188
activity	0.487	-0.514	-0.204	-0.472	0.363	-0.121	0.030	-0.269	**1**	**-1**	-0.048	-0.159
descriptivity	-0.487	0.514	0.204	0.472	-0.363	0.121	-0.030	-0.269	**-1**	**1**	0.048	0.159
TC	0.222	0.081	0.023	0.076	-0.312	0.254	0.431	-0.056	-0.048	0.048	**1**	0.400
STC	-0.154	0.321	0.366	-0.278	-0.250	0.480	0.499	0.188	-0.159	0.159	0.400	**1**

According to the bold-type figures in Table 5 and Cohen's (1977) correlation classification [32], the h-point strongly correlates with TTR and R_1, as does TTR with the h-point, R_1, and RR_{mc}. A strong correlation exists between R_1 and h-point, TTR, RR_{mc}, and the writer's view. Likewise, we can find a strong correlation between (1) RR_{mc} and TTR, R_1, RR, and writer's view, and (2) writer's view, R_1, and RR_{mc}. The strong correlation between quantitative indexes indicates an overlap in some information or a contribution to the author's writing style in homogeneous academic texts. Therefore, it is necessary to select one of them as the parameter index to calculate the text style. According to this principle, the twelve quantitative indexes can be divided into five independent unrelated or less relevant groups: (1) h-point, entropy, writer's view, verb distances, activity, thematic concentration, and secondary thematic concentration; (2) TTR, entropy, RR, verb distances, descriptivity, thematic concentration, and secondary thematic concentration; (3) R_1, entropy, verb distances, RR, activity, thematic concentration, and secondary thematic concentration; (4) RR_{mc}, h-point, entropy, verb distances, descriptivity, thematic concentration, and secondary thematic concentration; and (5) writer's view, h-point, entropy, verb distances, descriptivity, thematic concentration, and secondary thematic concentration.

4.1.3. Results of the Grouping Text Indexes through Principal Component Analysis

According to the grouping of quantitative indices following Pearson correlation coefficient analysis, each group still comprises seven text indices. There are still too many indicators for the text style analysis. The study conducted a five-time principal component analysis to reduce the text indices. Tables 6 and 7 present the results of Group 1.

Table 6. Total Variance of Group One Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	**2.453**	**35.037**	35.037	2.453	35.037	35.037	2.083	29.757	29.757
2	**1.661**	**23.722**	58.759	1.661	23.722	58.759	1.775	25.357	55.114
3	**1.408**	**20.113**	78.872	1.408	20.113	**78.872**	1.663	23.758	78.872
4	.705	10.070	88.942						
5	.557	7.959	96.900						
6	.165	2.363	99.263						
7	.052	.737	100.000						

Extraction Method: Principal Component Analysis.

Table 7. Component Matrix of Group One

	Component		
	1	2	3
h-point	-.494	.514	.662
Entropy	-.717	-.435	.265
Writer's view	**.914**	-.132	.074
Verb Distances	.571	.402	-.125
Activity	.110	**.938**	.162
Thematic Concentration	.312	-.245	**.847**
Secondary Thematic Concentration	.652	-.297	.368

Extraction Method: Principal Component Analysis.

a. three components extracted.

It can be seen from the initial eigenvalues in Table 6 that the coefficients of the three components are more significant than "1", and their cumulative contribution value reaches 78.872%. Table 7 indicates that among the seven quantitative indexes, the writer's view primarily explains the first component, with a coefficient value of 0.914; similarly, activity primarily explains the second component, with a coefficient value of 0.938. Thematic concentration mainly explains the third component, with an initial eigenvalue of 0.847. Through the principal component analysis method, the first group of the original seven quantitative indexes (through which the writer's view, activity, and thematic concentration can explain and represent most of the writing styles of homogeneous texts) plays a role in dimension reduction.

In this way, this study performed another four-time principal component analysis through SPSS 23 and obtained the other four groups' principal components: (2) TTR, verb distances, and thematic concentration; (3) R_1, verb distances, and thematic concentration; (4) entropy, descriptivity, and thematic concentration; and (5) entropy, descriptivity, and thematic concentration. Groups (4) and (5) were the same.

Through the use of the Pearson correlation coefficient and principal component analysis, the original twelve text indices were reduced to four groups of index combinations, each comprising three indices. In other words, the combination of (1) writer's view, activity, and thematic concentration; (2) TTR, verb distances, and thematic concentration; (3) R_1, verb distances, and thematic concentration; and (4) entropy, descriptivity, and thematic concentration can represent the writing style of the author in EFL homogeneous academic texts.

4.1.4. Analysis of the Stability of the Principal Text Quantitative Indexes

The five groups of quantitative text indexes representing the writing style of an author of EFL homogeneous academic texts involve eight indicators: *thematic concentration, verb distances, entropy, descriptivity, writer's view, activity, TTR, and richness*. The broken line chart illustrates the trend of stability for the primary eight text indices, with relevant values calculated. The specific results are shown in Figure 1 and Table 8.

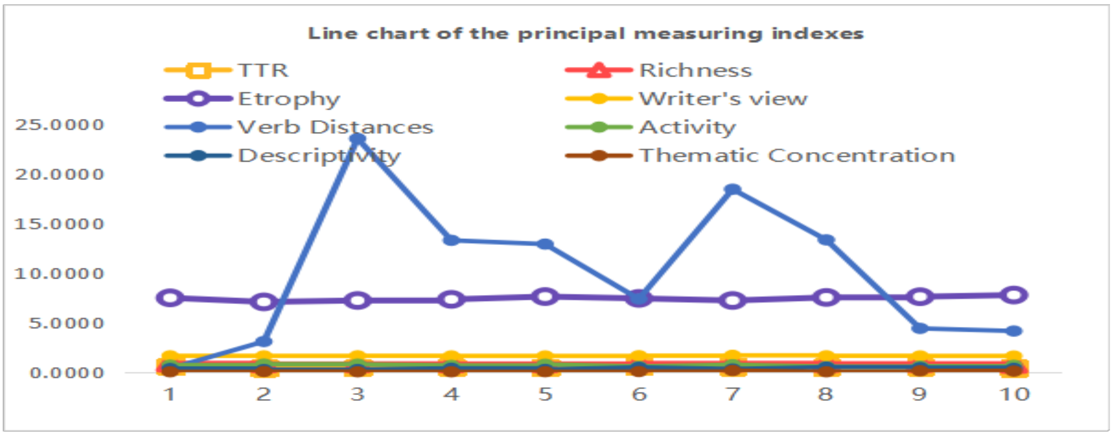

Figure 1. **The line chart trend of the principal measuring indexes**

Table 8. The main indexes' maximum, minimum, average, variance, and standard deviation

	Thematic Concentration	Verb Distances	Entropy	Descriptivity	Writer's View	Activity	TTR	Richness
	0.125	23.479	7.728	0.471	1.657	0.774	0.556	0.866
Min	0	0.341	7.037	0.226	1.589	0.529	0.385	0.737
Mean	0.0356	10.043	7.3868	0.336	1.609	0.664	0.4735	0.8199
Variance	0.02	55.170	0.045	0.008	0.0003	0.008	0.0034	0.0015
Sdev	0.0458	7.428	0.214	0.092	0.019	0.092	0.058	0.0385

Figure 1 clearly shows that the broken line chart (except for the one representing verb distances) of the eight text indexes is almost a straight line with no significant fluctuations. Only the verb distance data fluctuates considerably, with apparent peaks and troughs. Table 8 presents the stable characteristics of the data results for the eight indicators more accurately in terms of quantity. Except for verb distances, the maximum and minimum values of the other seven indicators are insignificant, the variance is far less than 1, and the standard deviation is significantly lower than one standard deviation. Therefore, except for verb distances, the overall data results for the main quantitative indexes of homogeneous texts remain relatively stable. The stability of all the main text indexes except verb distances verifies the hypothesis that the writing style of the same author in the same genre (English academic texts) in a specific period with similar text lengths (homogeneous texts) remains stable.

4.1.5. Validation of the Stability of Quantitative Indexes of the Homogeneous Texts

To further improve the reliability and validity of this study, 13 academic papers from the other three EFL teachers (six from Liu, two from Shi, and five from Wen) were selected for verification and analysis. The basic information of these 13 papers is shown in Table 3. As shown in Table 3, all papers are of an academic nature. Although the length of the articles varies, because the author and genre are the same, the parts still belong to EFL homogeneous texts. Additionally, all papers were written in English and published in various sources, spanning a 19-year period from 1997 to 2016. Compared to the ten texts mentioned earlier, the verification texts exhibit varying lengths and extended writing periods.

To meet the actual needs for verification, the authors first converted the 13 academic papers into Word documents. They then arranged them into 13 texts by removing charts, titles, abstracts, and references. Due to the complexity of the part-of-speech tagging process, this verification did not implement part-of-speech tagging for these papers. Then, the introductions and the conclusions from all 13 papers formed the homogeneous texts. All 26 homogeneous academic texts were imported into QUITA for calculation; the principal measuring indices, excluding part-of-speech tagging, are presented in the tables below (Tables 9 and 10).

Table 9. The data of the principal text indexes of the homogeneous texts (introductions)

Articles	TTR	Entropy	R1	RR	RRmc	Writer’s View
Liu’s Article 1	0.535424	6.146354	0.882062	0.018737	0.922974	1.808502
Liu’s Article 2	0.542406	6.414561	0.845188	0.017864	0.902242	1.890303
Liu’s Article 3	0.553719	6.577532	0.861241	0.016051	0.915884	1.803091
Liu’s Article 4	0.566774	6.295932	0.859382	0.019887	0.900309	1.848138
Liu’s Article 5	0.572514	6.689142	0.845312	0.017324	0.911931	1.814023
Liu’s Article 6	0.568387	6.284372	0.878282	0.017459	0.903244	1.926255
Shi’s Article 1	0.510871	5.953957	0.788043	0.026229	0.934426	2.091201
Shi’s Article 2	0.515385	6.012035	0.792623	0.020106	0.947758	1.988668
Wen’s Article 1	0.610204	5.715564	0.809902	0.027535	0.942946	2.616821
Wen’s Article 2	0.654321	5.907229	0.823529	0.025351	0.947366	2.074423
Wen’s Article 3	0.663866	5.654495	0.827161	0.022571	0.977492	2.240378
Wen’s Article 4	0.674142	5.725033	0.833077	0.023228	0.967332	2.394055
Wen’s Article 5	0.602278	5.801096	0.886076	0.028996	0.953326	2.614738

Table 9 shows that the data of six quantitative text indexes directly calculated by QUITA without part-of-speech tagging in the 13 academic papers of the three professors’ introduction homogeneous texts tend to be average and stable. Except for the limited index data of specific papers, they are almost even. The length of the text still has a particular impact on the results of the measuring index data; however, the data of the six indexes remain relatively stable for each of the professors. The stability of the main indexes from English academic introductions of homogeneous texts reflects the stability of each professor’s English writing style within a specific period.

Table 10. The data of the principal text indexes of the homogeneous texts (conclusions)

Articles	TTR	Entropy	R1	RR	RRmc	Writer's View
Liu's Article 1	0.537143	6.798806	0.798789	0.013184	0.918106	1.960689
Liu's Article 2	0.579505	6.869605	0.855565	0.013747	0.957521	2.180008
Liu's Article 3	0.500004	6.642384	0.803629	0.017419	0.943828	2.094355
Liu's Article 4	0.520161	6.454532	0.838712	0.018212	0.948572	2.075072
Liu's Article 5	0.580328	6.806981	0.798361	0.017253	0.939246	2.010968
Liu's Article 6	0.590674	6.622624	0.888956	0.014375	0.966557	2.036461
Shi's Article 1	0.647059	5.850342	0.827731	0.026057	0.946433	1.984297
Shi's Article 2	0.659545	5.920862	0.845962	0.022792	0.942335	1.876816
Wen's Article 1	0.716252	5.318131	0.904062	0.030062	0.949755	2.051136
Wen's Article 2	0.739583	5.978039	0.938802	0.028446	0.980554	2.981217
Wen's Article 3	0.779221	5.760039	0.928571	0.021083	0.981514	2.713018
Wen's Article 4	0.746429	5.948081	0.896116	0.022481	0.958604	2.403778
Wen's Article 5	0.765642	5.987436	0.918282	0.028016	0.945703	2.105266

Table 10 confirms that the data from the six indexes of the homogeneous texts of the conclusions of the 13 academic papers remain comparatively stable, with the differences between the maximum and minimum values of all the indicators not significant; the variance is far less than "1", and the standard deviation being also significantly lower than one standard deviation, which implies the stable writing style of the English academic homogeneous texts of the academic papers' bodies to each writer.

To obtain more compelling data, the researchers compared the authors' average data in the same quantitative indexes, including writer's view, RRmc, RR, R_1, entropy, and TTR from the introductions. Figure 2 shows the differences.

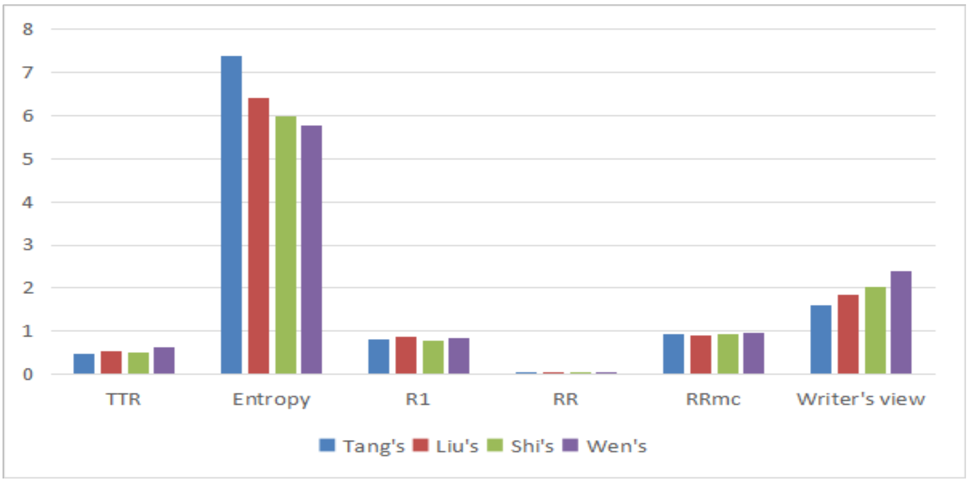

Figure 2. **The comparison of four authors' quantitative indexes of homogeneous texts (Introductions)**

Figure 2 illustrates the overall differences among the four authors' text indexes within the same text genres (Introductions), despite the similar data in some indexes. It further demonstrates that different authors have their own writing style within a specific period, with each author maintaining relatively stable quantitative indexes.

4.2. Discussion

The data of the quantitative text indexes of the homogeneous academic texts are the quantitative representation of the writing style of the same author of the homogeneous academic texts. Whether the indicator result data maintains overall stability reflects the author's writing style trend within a specific period. The results show that, except for verb distances, the principal measuring indices for EFL homogeneous academic texts are relatively stable. This stability reflects the author's writing style to some extent, and different EFL writers show individual differences in academic writing. After all, English is a foreign language for the authors involved in this paper. After many years of English acquisition, an EFL learner's vocabulary, grammatical framework, and writing style stabilize. It is difficult to change them at a specific stage, founded by Selinker's (1972) interlanguage theory, especially the fossilization theory [33]. Fossilization refers to the point at which L2 learners reach a plateau in their language development, where their abilities cease to progress. Fossilization happens at all levels, including writing ability. In this study, the EFL writers' academic writing abilities are unlikely to reach the level of native speakers and maintain their writing style.

Most quantitative indices of the EFL homogeneous academic texts are relatively stable, but the verb distances fluctuate considerably. Verbs are content words that are highly sensitive to factors beyond authorship, including genre, topic, and addressee [34]. Besides, this result is directly related to the invariant properties of quantitative text indexes shown in Table 11.

Table 11. The attribute characteristics of the measuring indexes of texts

Measuring indexes	Vocabulary	Vocabulary+	linear	Two-sample test	Length of texts
h-point	√	×	-	×	Extremely strong+
TTR	√	×	-	×	Strong -
R_1	√	×	-	√	Strong-
RR	√	×	-	√	Moderate -
RR_{mc}	√	×	-	×	Moderate -
entropy	√	×	-	√	Strong +
verb distances	×	√	+	×	Extremely weak/irrelevant
activity	×	√	-	√	Weak+
descriptivity	×	√	-	√	Weak -
writer's view	×	√	-	×	Moderate -
TC	×	√	-	√	Weak -
STC	×	√	-	√	Weak-

Table 11 intuitively reflects the dichotomy of text index characteristics, that is, whether it belongs to the lexical level, is linear, is suitable for sample testing, and is related to text length. For example, some indices are closely related to text lengths, such as h-point, TTR, and R1. Verb distances belong to the non-lexical level (phrase, clause, sentence), which have linear relationships and are unrelated to the text length. In other words, verbs (including gerunds, participles, the original form of verbs, and verb forms under each tense voice) cannot change their positions at will. If the author changes their position, the structure of sentences related to the verbs involved will change. Given distinct themes, the author holds varied views, intentions, preferences, and tendencies, with a solid subjective mood that reflects different tenses. These changes and choices are realized through the change and selection of verbs. Therefore, even if it is the same genre of texts with different themes, the verb distances also vary, exhibiting unstable characteristics. Compared with verb distances, other indexes are non-linear; that is to say, they are not affected by the choice of words and sentences, and changes in their position have no fundamental impact.

Of course, the stability of the measuring index may also be related to relevant factors involved in the calculation formula and others, such as the proportion of quotations added to the text, which, due to the limited space in this paper, will not be repeated in detail. The stability of the quantitative text indexes is a quantitative reflection of the stability of the text writing style. Indexes other than verb distances can be selected to detect EFL homogeneous academic texts, such as (1) the writer's view, activity, and thematic concentration, and (2)entropy, descriptivity, and thematic concentration. The second combination, which frequently appears, can primarily be used as a leading indicator to judge whether the EFL homogeneous academic texts are stable or not in terms of writing style.

According to the validation process, the data of the three professors' measuring indexes appear to be convergent, especially regarding R_1, RR, and $RR_{mc.}$ It partly results from the fact that most papers have been revised by incorporating the relevant suggestions of reviewers or editors, which means that more than one author may contribute to these papers. It further confirms that writers with different personalities exhibit individual differences in their writing styles. Another factor that these indexes' low absolute values can account for is the trend in that low values can hardly reflect the differences and fluctuations.

5. Verified hypotheses and their application to academic ethics education in university libraries

Thus far, it has been shown that the data of all the quantitative indexes except those of verb distances remain stable. The text indexes are the quantitative representation of the author's writing style. In this way, the hypothesis that the writing style of the same author's EFL homogeneous academic texts remains stable has been verified. The first two questions have also been answered. All the quantitative indices, except for verb distances, representing the author's writing style, remain stable through their stable data. Four groups of text indexes can represent the writing style of the same author. They are (1) the writer's view, activity, and thematic concentration; (2) TTR, verb distances, and thematic concentration; (3) R_1, verb distances, and thematic concentration; and (4) entropy, descriptivity, and thematic concentration. Verb distance instability is considered, and the first and fourth groups are preferred for practical and economic purposes.

Libraries are also the academic center of a university, where teachers and students can find almost all the academic materials they need. Additionally, libraries are ideal places for implementing academic ethics education. For the first class, academic writing teachers in Western countries will likely take their students to the library. They emphasize the importance of academic ethics by teaching academic writers to avoid academic misconduct and adhere to established academic standards and guidelines. Libraries play a role in academic ethics education. However, universities and colleges in China often overlook academic ethics education to some degree for unknown reasons. Likewise, the academic ethics education of libraries needs strengthening.

The verified hypothesis that the writing style of the same author's EFL homogeneous academic texts remains stable within a specific period has practical implications for academic ethics education in libraries, as it facilitates authorship attribution. Obtaining homogeneous writing texts for EFL academic papers and theses is relatively easy. For an EFL academic paper, the other homogeneous texts can be parts of other papers from the same author, provided all the papers have no coauthor(s). Additionally, for the same EFL academic paper, homogeneous texts can be obtained within the same paper in different drafts. For example, the introduction, literature review, theoretical framework, methods, results, discussion, and conclusion from the same paper can form homogeneous texts according to the first, second, and final drafts. As far as EFL academic theses are concerned, obtaining homogeneous texts is similar to that of EFL academic papers. On the one hand, the different versions of these from the same author can be called homogeneous academic texts. On the other hand, parts with similar paragraph lengths in the same thesis (e.g., introduction, literature review, theoretical framework, research questions, methods, results, discussion, limitations, conclusions, and acknowledgments) can also be considered homogeneous academic texts.

Once the homogeneous texts are identified, authorship can be easily and accurately determined by importing the homogeneous texts into QUITA and comparing the data of representative quantitative indexes. To be more specific, the general procedure to resolve cases of unknown authorship can follow Grieve's (2007) five steps [35]. The first step is to identify a valid set of possible authors by analyzing the quantitative indices of the unknown texts. The second step is to establish a corpus of potential authors by compiling EFL academic writings. Third, test all possible texts through QUITA and collect index data from the writer's perspective, including activity, thematic concentration, entropy, and descriptivity. Next, focus on the possible authors' stock index data and narrow the range of authors. Finally, try to find the best writings in the corpus that match the disputed writing.

Regarding the libraries, librarians are required to download the QUITA software and install it on each computer in the libraries. Then, all published and unpublished papers and theses from university and college teachers and students are encouraged to be sent to the librarians, so that an academic writing corpus can be constructed. In this way, the teacher can promote academic ethics education in libraries by guiding students to adopt a writing style that avoids misconduct.

The best way for EFL learners to avoid academic misconduct in writing is to automatically detect the texts through the widely recognized duplicate and novelty search software. However, it is not easily accessible for university students to apply the software. Besides, it, in most cases, costs a sum of money. This study can contribute to the academic ethics education in libraries in higher education in non-English speaking countries. Explicit instruction in applying homogeneous text recognition can help writers, editors, tutors, and others automatically form a habit of checking the stability of quantitative writing indexes to avoid academic misconduct.

6. Conclusions

The research proposes a new method of combining text analysis with an author's writing style. Utilizing the stability of text indexes of homogeneous texts provides a novel approach for detecting unknown texts in English academic theses and papers, as well as EFL academic writing misconduct. However, in terms of academic rigor, there are still some limitations in this study: (1) only one genre of homogeneous texts has been studied in this paper, and the number of homogeneous texts is also quite limited, and (2) the number of research authors is limited. From the perspective of reliability and validity, an increase in the repeatability and frequency of the study is needed; (3) Further, the stability analysis and application of the measuring indexes of homogeneous writing stay only at the theoretical level, which calls for more practical testing; (4) the requirements of English academic homogeneous texts are not unequivocal. For example, there has yet to be a consensus on the length of homogeneous texts and their period; and (5) the four authors in this study come from the same country, which needs improving by researching more authors from more non-English speaking countries. These limitations also serve as the basis and direction for future studies.

A small, closed set of candidate authors and unlimited training texts for each still pose challenges to the researchers involved. Furthermore, homogeneous writing reflects the author's writing style and thinking within a specific period. Quantitative indices specially used for text analysis can detect this recessive feature. The data results represent the author's writing style and personality in a quantitative manner. Although this study still has some limitations, the results can provide potential convenience for most tutors, editors, and even authors and readers through this new method of author attribution. We hope this new approach can help regulate academic writing behavior and promote conscious awareness of compliance with writing ethics. Besides, language planning in higher education should concern the self-checking before publishing academic writing.

Acknowledgement

This study was conducted and supported by the China Association for Higher Education Planning Program with ID 24WY0405. Any views, findings, suggestions, or conclusions written in this article are those of the authors and do not necessarily reflect the views of the fund.

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